

Split-Plot Design

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Motivation

- **Why do we need split-plot designs?**

Some factors are:

Hard to change (expensive, slow, physical constraints)

Easy to change

We cannot fully randomize all factors in the same way

- Two layers of experimental units:
- **Whole plots** → receive hard-to-change factor
- **Subplots** → receive easy-to-change factor

Example 1: Industrial process with furnace setting and cooling method

- An engineer studies product strength under: **Furnace temperature:** 800°C, 900°C and **Cooling method:** air, water, oil
- Response: final material strength
- Changing furnace temperature is hard: furnace must be reconfigured, temperature stabilization takes time
- Changing cooling method is easy: after heating, different specimens can be cooled in different ways
- **whole plot factor** = furnace temperature
- **subplot factor** = cooling method

Example 2: Agricultural irrigation and fertilizer

- A researcher wants to study crop yield as a function of: **Irrigation level**: low, medium, high, **Fertilizer type**: A, B, C, D
- Response: yield per subplot
- Changing **irrigation level** is hard: irrigation equipment is installed at the field level, it is difficult to supply different irrigation levels to tiny adjacent plots
- Changing **fertilizer type** is easy: fertilizer can be applied separately to smaller subplots
- whole plot factor = irrigation
- subplot factor = fertilizer

Structure of Split-Plot Design

Level	Unit	Factor
Whole plot	large unit	Factor A (hard)
Subplot	small unit	Factor B (easy)

- Each whole plot gets one A level
Within each whole plot, all B levels are applied
- For example, if Factor A has 2 levels, Factor B has 3 levels, and we have 4 whole plots, then two whole plots get low level of A, two whole plots get high level of A, and within each whole plot, all B levels are applied.

Design table

Furnace Run	Temperature	Cooling Method
1	800°C	Air
1	800°C	Water
1	800°C	Oil
2	900°C	Air
2	900°C	Water
2	900°C	Oil
3	800°C	Air
3	800°C	Water
3	800°C	Oil
4	900°C	Air
4	900°C	Water
4	900°C	Oil

Modeling

- Observations in same whole plot are correlated

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + w_k + \epsilon_{ijk}$$

Components

- α_i : fixed effect of A
 - β_j : fixed effect of B
 - $w_k \sim N(0, \sigma_w^2)$: whole-plot random effect
 - $\epsilon_{ijk} \sim N(0, \sigma^2)$: subplot error
- Two Sources of Error

Covariance Structure

- Variance of each observation:

$$\text{Var}(Y) = \sigma_w^2 + \sigma^2$$

- Same whole plot:

$$\text{Cov}(Y_i, Y_j) = \sigma_w^2$$

- Different whole plots:

$$\text{Cov}(Y_i, Y_j) = 0$$

- Observations within a whole plot are **correlated**

Matrix Form of Split-Plot Model

$$\mathbf{Y} = X\beta + Z\mathbf{w} + \epsilon$$

$$\text{Var}(\mathbf{Y}) = V = \sigma_w^2 Z Z^\top + \sigma^2 I$$

Fixed effects

$$\beta = \begin{bmatrix} \mu \\ \alpha \\ \beta \\ \alpha\beta \end{bmatrix}$$

Random effects

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_K \end{bmatrix}, \quad \mathbf{w} \sim N(0, \sigma_w^2 I)$$

When to Use Split-Plot

- Some factors are hard to change
- Full randomization is impractical: restricted randomization
- Natural hierarchical structure exists